PROBLEM DEFINITION:

Version 1:

~~VI.~~ IP pipeline:

Input: image (portrait / landscape)

Output: Processed output (Clean, transformed image)

→ Constraints:

Input: image (jpg / png)

Size: for now any size / quality.

Technical: CPU run.

constraints Use Python libraries.

Basic Performance constraint.

PIPELINE:

1) Input image: Take the input from the user (Version 1: ~~Dro~~ Select from folder type)

2) Grayscale: Convert the image to grayscale.

3) Resize: Ask the user to resize according to how much does he need. in that photo

4) <u>Preprocessing</u>: Noise reduction techniques, modifying the image, to get a better quality image.

5) <u>Save outputs</u>: Save the image in a specified folder (Version 1).

src/ directory changes:

→ src/

- ├ __init__.py
- ├ pipeline.py → Processing flow / pipeline
- ├ processing.py → Modules / operations for Image processing
- ├ utils → Load / save ops.
- ├ config.py → constants / parameters.
- └ main.py → Entry point

<u>Concepts learnt today</u>:

1. Pipeline Thinking: Connection of various independent modules into a single workflow so that for a user, ~~the~~ he / she is only concern is i/p. (Abstraction of modules)

2. Single responsibility principle : a class / module should have only a single, cohesive task.

3. Interface contract : An agreement to define the expected behaviour of inputs / outputs between 2 modules / components, without specifying internal implem.

4. Constraint awareness : Identify and manage project limitations in terms of budget, time, performance, tech. feasibility etc.

5. Refactoring : Improve and design the structure of code and directories without affecting the external behaviour.